

PCA

| $x$                                                                   | $y$ | $(x - \bar{x})$ | $(y - \bar{y})$ | $(x - \bar{x})(y - \bar{y})$ | $(x - \bar{x})^2$ | $(y - \bar{y})^2$ |
|-----------------------------------------------------------------------|-----|-----------------|-----------------|------------------------------|-------------------|-------------------|
| 2.5                                                                   | 2.4 | 0.69            | 0.49            | 0.3381                       | 0.4761            | 0.2401            |
| 0.5                                                                   | 0.7 | -1.31           | -1.21           | 1.5851                       | 1.7161            | 1.4641            |
| 2.2                                                                   | 2.9 | 0.39            | 0.99            | 0.3861                       | 0.1521            | 0.9801            |
| 1.9                                                                   | 2.2 | 0.09            | 0.29            | 0.0261                       | 0.0081            | 0.0841            |
| 3.1                                                                   | 3.0 | 1.29            | 1.09            | 1.4061                       | 1.6641            | 1.1881            |
| 2.3                                                                   | 2.7 | 0.49            | 0.79            | 0.3871                       | 0.2401            | 0.6241            |
| 2                                                                     | 1.6 | 0.19            | -0.31           | -0.0589                      | 0.0361            | 0.0961            |
| 1                                                                     | 1.1 | -0.81           | -0.81           | 0.6561                       | 0.6561            | 0.6561            |
| 1.5                                                                   | 1.6 | -0.31           | -0.31           | 0.0961                       | 0.0961            | 0.0961            |
| 1.1                                                                   | 0.9 | -0.71           | -1.01           | 0.7171                       | 0.5041            | 1.0201            |
| <u><math>\bar{x} = 1.81</math></u> <u><math>\bar{y} = 1.91</math></u> |     |                 |                 | <u>5.539</u>                 | <u>5.549</u>      | <u>6.449</u>      |

# Covariance Matrix

$$C = \begin{bmatrix} \overset{0.617}{\text{cov}(x, x)} & \overset{0.6154}{\text{cov}(x, y)} \\ \underset{0.6154}{\text{cov}(y, x)} & \underset{0.716}{\text{cov}(y, y)} \end{bmatrix}$$

$$\frac{5.549}{9} \quad \text{cov}(x, x) = \frac{\sum (x - \bar{x})(x - \bar{x})}{n-1} \quad \boxed{0.617}$$

$$\frac{5.539}{9} \quad \text{cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1} \quad \boxed{0.6154}$$

$$\frac{6.449}{9} \quad \text{cov}(y, y) = \frac{\sum (y - \bar{y})(y - \bar{y})}{n-1} \quad \boxed{0.716}$$

Step 2 = Calculate Eigen values.

$$|C - \lambda I| = 0$$

$$\left| \begin{bmatrix} 0.617 & 0.6154 \\ 0.6154 & 0.716 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 0.617 & 0.6154 \\ 0.6154 & 0.716 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$\left| \begin{bmatrix} 0.617 - \lambda & 0.6154 \\ 0.6154 & 0.716 - \lambda \end{bmatrix} \right|$$

$$(0.617 - \lambda)(0.716 - \lambda) - (0.6154)^2$$

$$\lambda^2 - (0.617\lambda) - 0.716\lambda - (0.378)$$

$$\boxed{\lambda^2 - 1.33\lambda + 0.063 = 0} \quad +0.4417$$

$$\underline{\lambda_1 = 1.2808}$$

$$\underline{\lambda_2 = 0.0491}$$

### Step 3 = Eigen vectors

$$CV_1 = \underline{N} V_1$$

$$\begin{bmatrix} \underline{0.617} & \underline{0.6154} \\ \underline{0.6154} & \underline{0.716} \end{bmatrix} * \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \underline{1.2808} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

$$0.617 x_1 + 0.6154 y_1 = 1.2808 x_1$$

$$0.617 x_1 - 1.2808 x_1 = -0.6154 y_1$$

$$\boxed{-0.6638 x_1 = -0.6154 y_1}$$

$$0.6154 x_1 + 0.716 y_1 = 1.2808 y_1$$

$$\begin{aligned} 0.6154 x_1 &= 1.2808 y_1 - 0.716 y_1 \\ &= \boxed{0.6154 x_1 = 0.6654 y_1} \end{aligned}$$

$$-0.6638x_1 = 0.615y_1$$

$$x_1 = 0.9270$$

$$v_1 = \begin{bmatrix} 0.927 \\ 1 \end{bmatrix} \quad \frac{0.927}{\sqrt{(0.927)^2 + (1)^2}}$$

$$v_1^* = \begin{bmatrix} 0.679 \\ 0.733 \end{bmatrix} = \frac{0.679}{\sqrt{(0.927)^2 + (1)^2}} \\ = 0.733$$

$$Cv_2 = N_2 v_2$$

$$\begin{bmatrix} 0.617 & 0.6154 \\ 0.6154 & 0.716 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = 0.0491 \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$0.617x_2 + 0.6154y_2 = 0.0491x_2$$

$$\begin{cases} 0.6154x_2 + 0.716y_2 = 0.0491y_2 \end{cases}$$

$$\rightarrow 0.617x_2 - 0.0491x_2 = -0.6154y_2$$

$$\boxed{0.5679x_2 = -0.6154y_2}$$

$$0.6154x_2 = 0.0491y_2 - 0.716y_2$$

$$0.6154x_2 = -0.6669y_2$$

$$0.5679x_2 = -0.6154y_2$$

$$x_2 = -0.6154/0.5679$$

$$x_2 = -1.0836$$

$$v_2 \begin{bmatrix} -1.0836 \\ 1 \end{bmatrix}$$

$$\frac{-1.0836}{\sqrt{-1.0836^2 + 1^2}}$$

$$v_2^* \begin{bmatrix} -0.7348 \\ 0.678 \end{bmatrix} = \frac{1}{\sqrt{-1.0836^2 + 1^2}}$$

| x   | y   |
|-----|-----|
| 2.5 | 2.4 |
| 0.5 | 0.7 |
| 2.2 | 2.9 |
| 1.9 | 2.2 |
| 3.1 | 3.0 |
| 2.3 | 2.7 |
| 2   | 1.6 |
| 1   | 1.1 |
| 1.5 | 1.6 |
| 1.1 | 0.9 |

$$\begin{bmatrix} 0.679 \\ 0.733 \end{bmatrix}$$

| Rf    | $\lambda_1$      |
|-------|------------------|
| 3.45  | $\boxed{1.2808}$ |
| 0.85  | $\lambda_2$      |
| 3.475 | $\lambda_2$      |
| 2.9   | $\lambda_1$      |
| 4.40  | $\lambda_1$      |
| 3.53  | $\lambda_2$      |
| 2.52  | $\lambda_2$      |
| 1.48  | $\lambda_1$      |
| 2.18  | $\lambda_2$      |
| 1.40  | $\lambda_1$      |

$$\lambda_1 \begin{bmatrix} 0.679 \\ 0.733 \end{bmatrix}$$

$$\lambda_2 \begin{bmatrix} -0.7348 \\ 0.678 \end{bmatrix}$$

# Moving Average.

|     |       | <u>MA3</u> | M.A5 | 7 | 5.5/3  |
|-----|-------|------------|------|---|--------|
| 2.5 | —     | —          | —    | — | —      |
| 0.5 | —     | —          | —    | — | —      |
| 2.2 | 1.833 | —          | —    | — | 4.15/3 |
| 1.9 | 1.53  | —          | —    | — | 7.2/3  |
| 3.1 | 2.4   | 2.04       | —    | — | 7.3/3  |
| 2.3 | 2.43  | —          | —    | — | 10.2/5 |
| 2   | —     | —          | —    | — | —      |
| 1   | —     | —          | —    | — | —      |
| 1.5 | —     | —          | —    | — | —      |
| 1.1 | —     | —          | —    | — | —      |